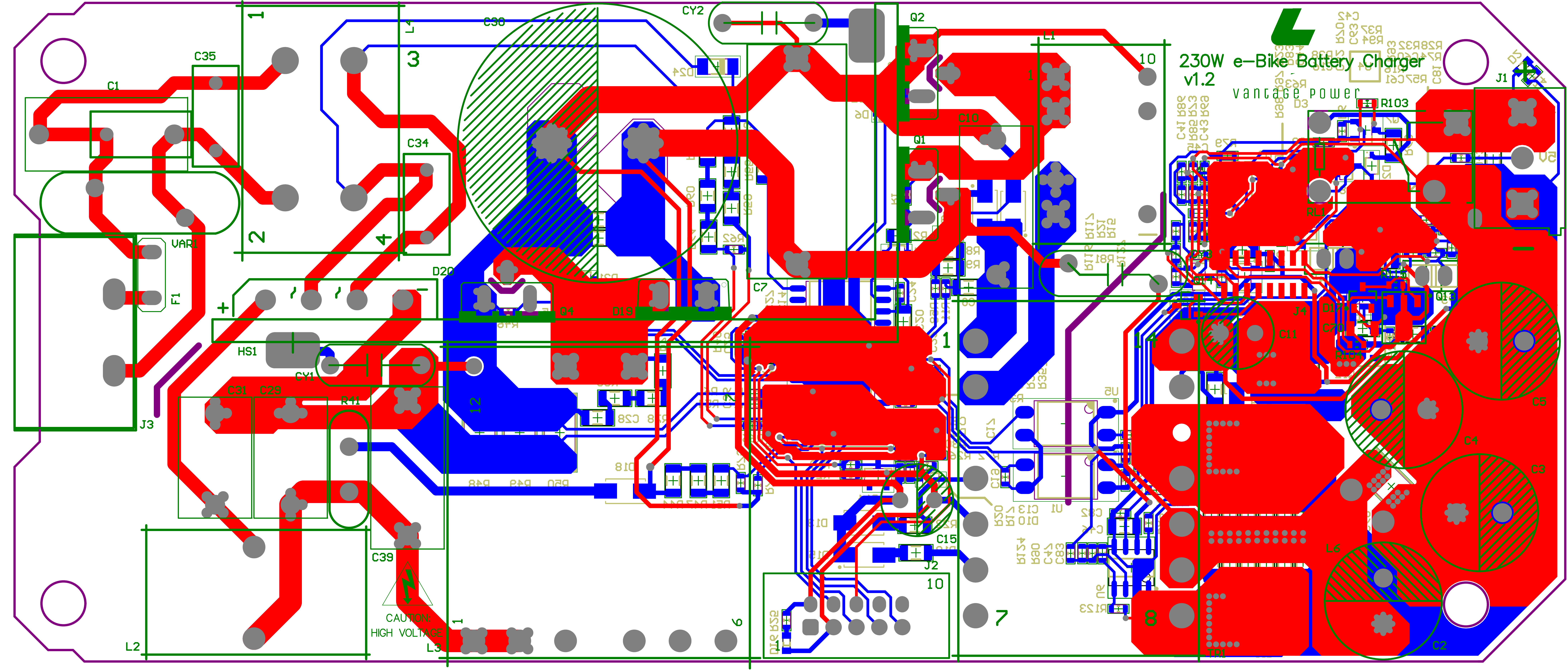
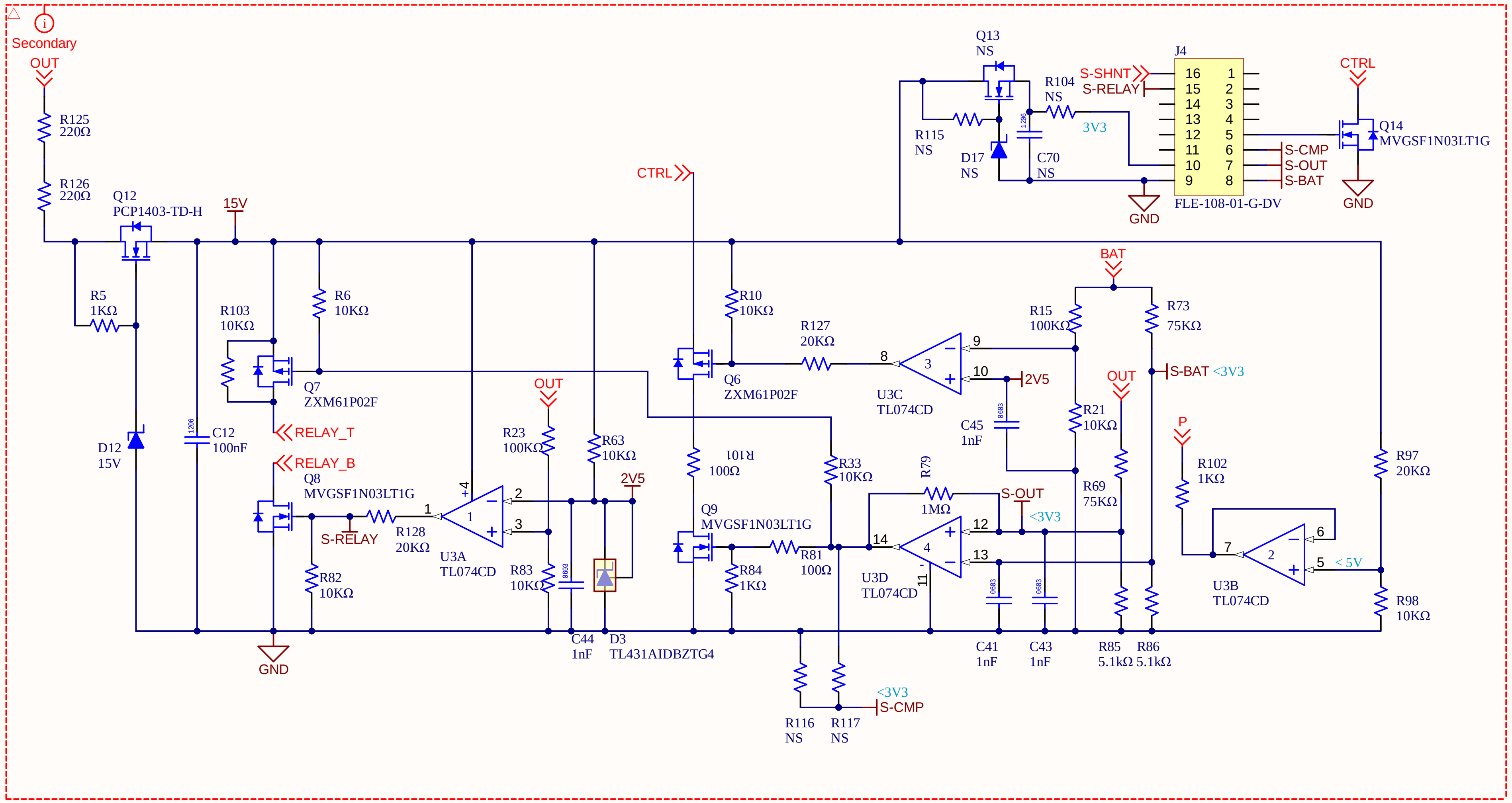

230W e-Bike Battery Charger
v1.2
vantage power
CAUTION
HIGH VOLTAGE
C1
C35
C34
C30
CY2
CY1
Q2
Q1
L1
L2
L3
L4
L6
J1
J2
J3
VAR1
F1
HS1
C31
C29
R41
C39
D20
D19
Q4
C7
C10
C15
C11
C2
C3
C4
C5
R103
Q13
RL1
TR1

REVISION	DESCRIPTION	DATE	APPROVED



APPROVALS	DATE
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DSN: *	--/--/--
CHK: *	--/--/--

REFERENCE DOCUMENTS

BOM:

ASSY DWG:

FAB DWG:

PCB DWG:

PROJECT *

Altium

ADDRESS 1
ADDRESS 2
ADDRESS 3
ADDRESS 4

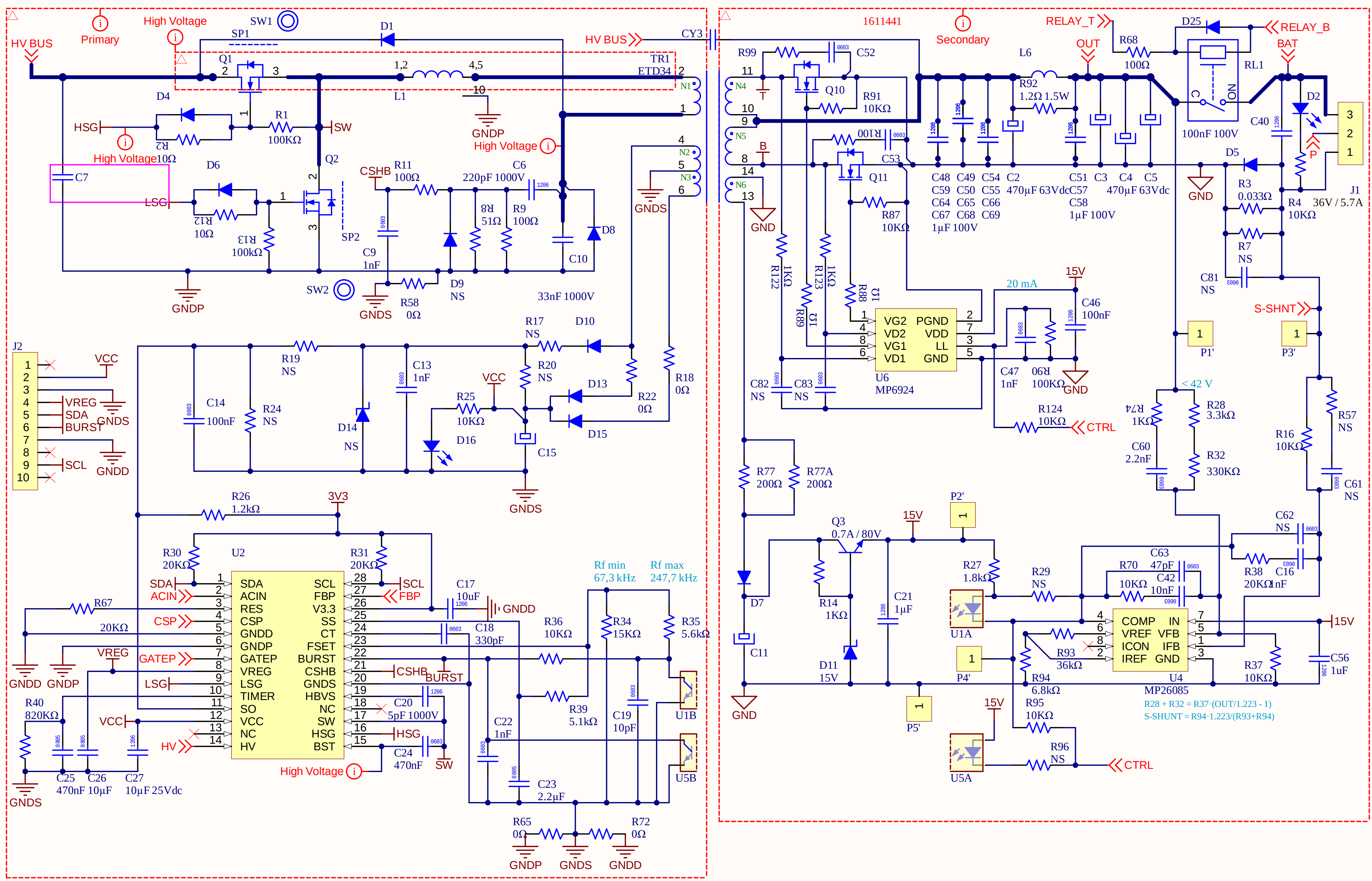
PROJECT REVISION: Not in version control
DOCUMENT REVISION: Not in version control
DESIGN ITEM:

TITLE *

SIZE B | CAGE CODE | DWG NO. | REV

SCALE: | FILE NAME CTRL.SchDoc | SHEET * OF *

REVISION	DESCRIPTION	DATE	APPROVED



APPROVALS	DATE
ENG: *	--/--/--
DSN: *	--/--/--
CHK: *	--/--/--

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*

SIZE	CAGE CODE	DWG NO.	REV
B			

SCALE: FILE NAME LLC.SchDoc SHEET * OF *

REVISION	DESCRIPTION	DATE	APPROVED

REV.	DESCRIPTION	DATE	AUTHOR
v1.0	Initial version of the board	Jun, 2018	Kaone Bogopa
v1.1	Changes in this version: 1.- Add a CM filter L5 2.- Remove EARTH ACIN sense 3.- Add PFC Snubber 4.- Syncronous rectification (MP6924) 5.- CCCV controller (MP26085) 6.- LC Output Filter L6 7.- Circuitry to implement a soft battery connection	Oct, 2018	Kaone Bogopa
v1.2	Changes in this version: 1.- Add C82, C83, C84. 2.- Add R77A R125, R126 to help with the power dissipation of the LDO. 3.- Add R127 and R128 to protect the MOSFETs. 4.- Increase C1 to pass EN55032 Class B.	Nov, 2019	Kaone Bogopa

PAGE Nº	DOCUMENT NAME
1	MAIN_005
2	PFC
3	LLC
4	CTRL

APPROVALS	DATE
ENG: Kaone Bogopa	--/--/--
DSN: *	--/--/--
CHK: *	--/--/--

REFERENCE DOCUMENTS
BOM:
ASSY DWG:
FAB DWG:
PCB DWG:

PROJECT *

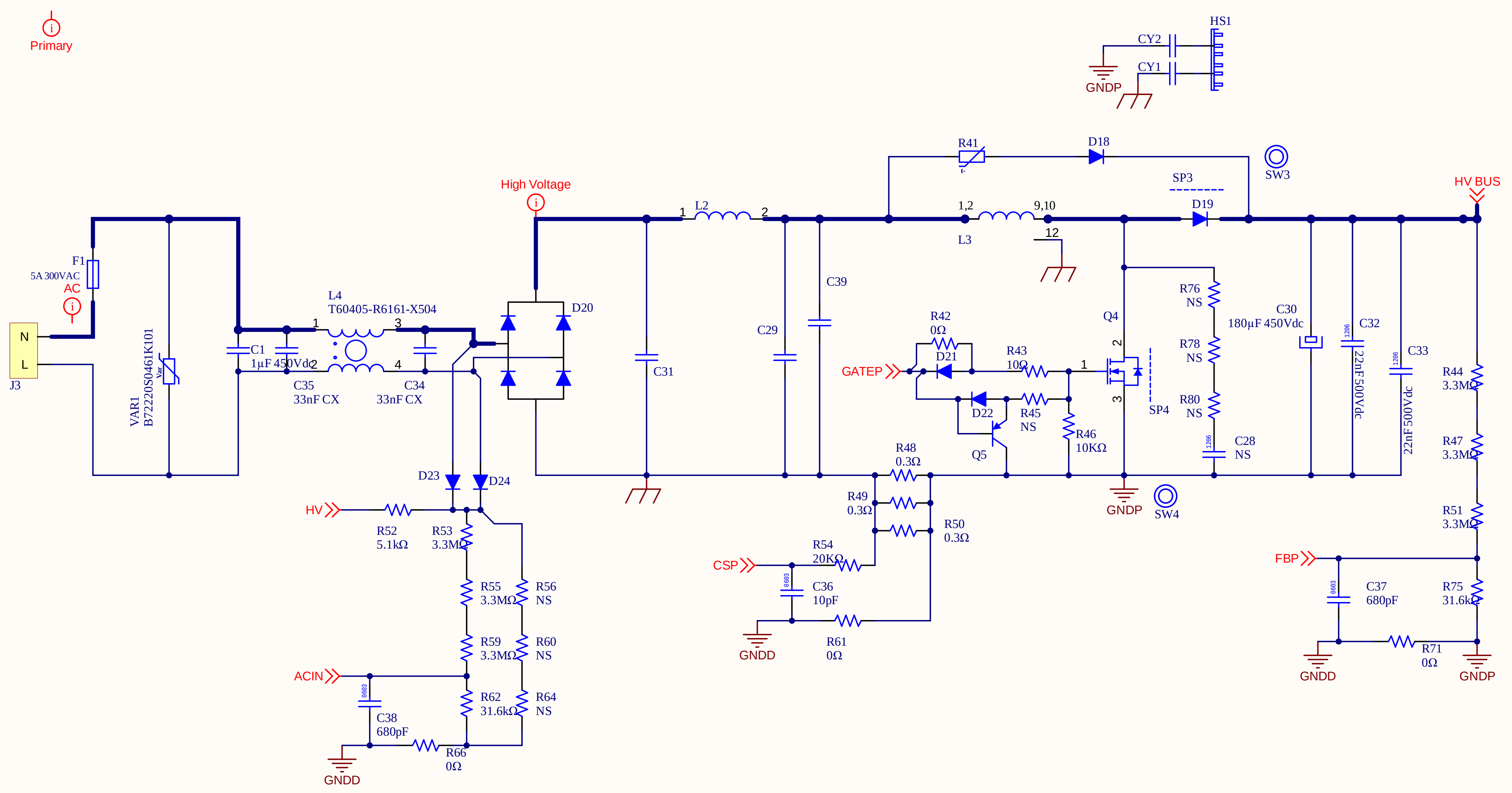
Altium
ADDRESS 1
ADDRESS 2
ADDRESS 3
ADDRESS 4

PROJECT REVISION: Not in version control
DOCUMENT REVISION: Not in version control
DESIGN ITEM:

TITLE
230W E-bike charger

SIZE B | CAGE CODE | DWG NO. | REV
SCALE: | FILE NAME Main_005.SchDoc | SHEET * OF *

REVISION	DESCRIPTION	DATE	APPROVED



APPROVALS	DATE
ENG: *	--/--/--
DSN: *	--/--/--
CHK: *	--/--/--

REFERENCE DOCUMENTS
BOM:
ASSY DWG:
FAB DWG:
PCB DWG:

PROJECT *

Altium

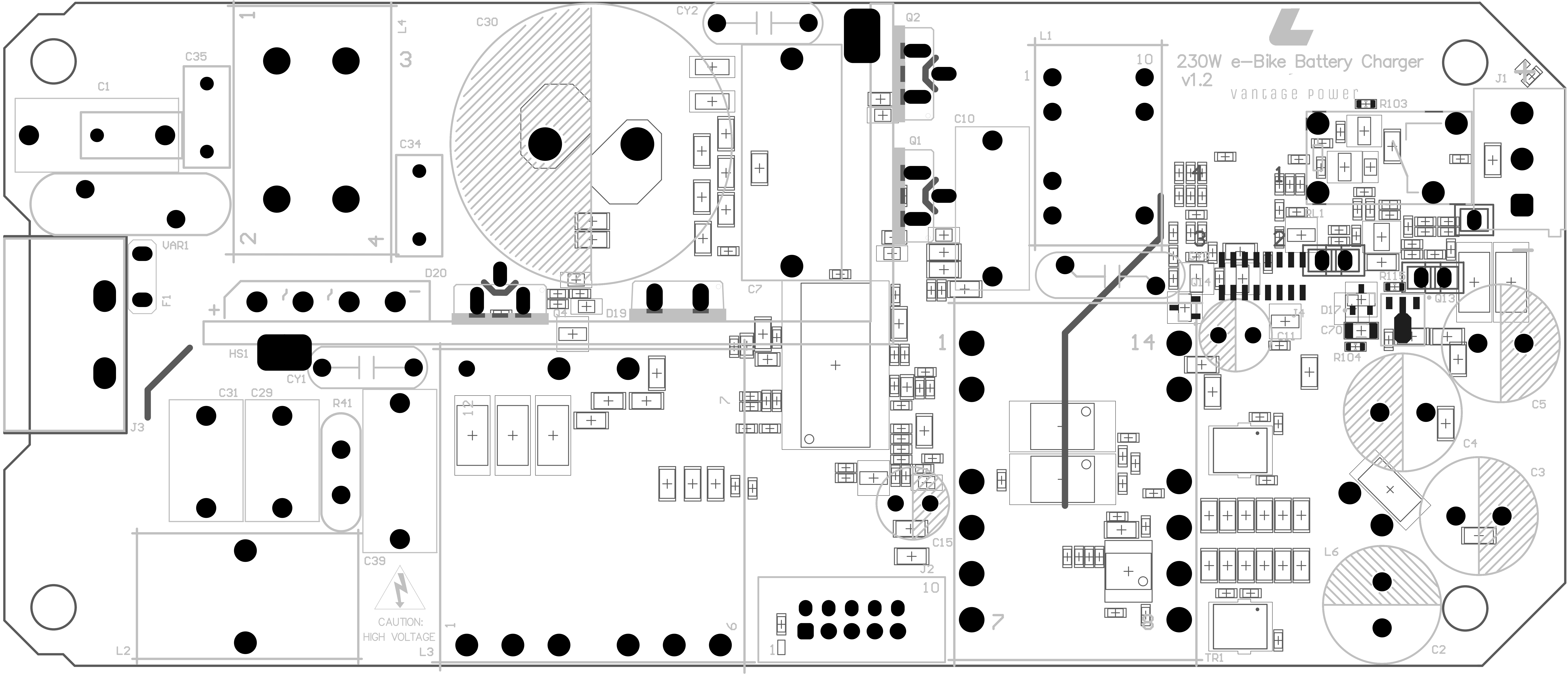
ADDRESS 1
ADDRESS 2
ADDRESS 3
ADDRESS 4

PROJECT REVISION: Not in version control
DOCUMENT REVISION: Not in version control
DESIGN ITEM:

TITLE *

SIZE B | CAGE CODE | DWG NO. | REV

SCALE: | FILE NAME PFC.SchDoc | SHEET * OF *


230W e-Bike Battery Charger
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vantage power
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C5
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C3
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L6
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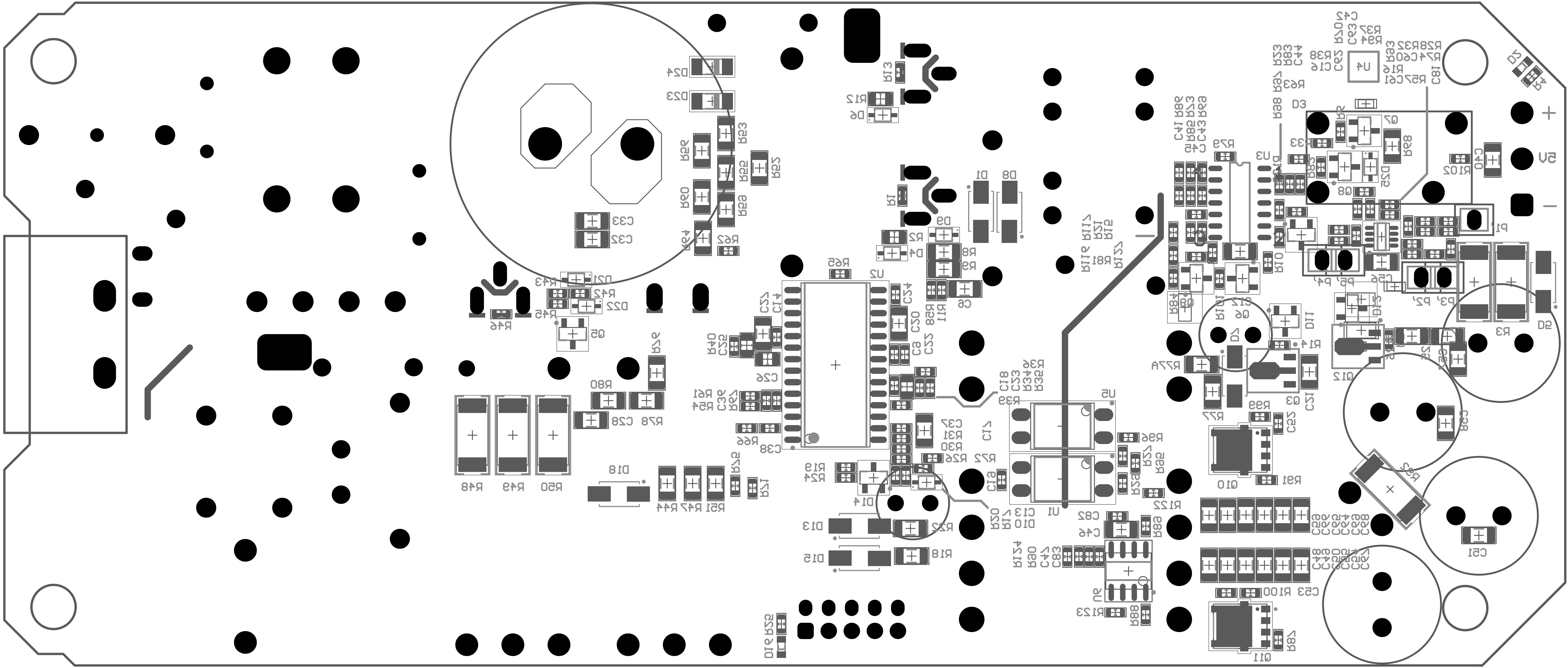

Symbol	Count	Hole Size	Plated	Hole Type	Drill Layer Pair	Via/Pad	Pad Shape	Template	Routed Path Length
G	160	0.300mm (11.81mil)	PTH	Round	Top Layer - Bottom Layer	Via	Rounded	v65h30	-
▣	21	0.400mm (15.75mil)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c80h40	-
□	95	0.500mm (19.69mil)	PTH	Round	Top Layer - Bottom Layer	(Mixed)	Rounded	(Mixed)	-
✚	4	0.800mm (31.50mil)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c180h80	-
○	7	0.950mm (37.40mil)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded	r160_225h95r80	-
⌧	29	1.000mm (39.37mil)	PTH	Round	Top Layer - Bottom Layer	Pad	(Mixed)	(Mixed)	-
▼	9	1.016mm (40.00mil)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded	(Mixed)	-
◎	10	1.100mm (43.31mil)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded	(Mixed)	-
D	2	1.200mm (47.24mil)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c250h120	-
✡	2	1.250mm (49.21mil)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded	r305_180h125r90	-
⌘	4	1.270mm (50.00mil)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c203h127	-
⌑	2	1.300mm (51.18mil)	NPTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c0hn130	-
⌑	28	1.300mm (51.18mil)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded	(Mixed)	-
⊕	7	1.400mm (55.12mil)	PTH	Round	Top Layer - Bottom Layer	Pad	(Mixed)	(Mixed)	-
C	4	1.500mm (59.06mil)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c300h150	-
B	2	1.500mm (59.06mil)	PTH	Slot	Top Layer - Bottom Layer	Pad	Rounded	r350_250h150_180r125	0.300mm (11.81mil)
F	1	1.600mm (62.99mil)	NPTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c0hn160	-
E	13	1.600mm (62.99mil)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded	(Mixed)	-
✡	2	2.000mm (78.74mil)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c370h200	-
▽	2	2.500mm (98.43mil)	PTH	Round	Top Layer - Bottom Layer	Pad	Rounded Rectangle	r600_400h250r100	-
A	2	3.200mm (125.98mil)	NPTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c0hn320	-
◇	4	4.900mm (192.91mil)	NPTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c0hn490	-
	410 Total								

Slot definitions : Routed Path Length = Calculated from tool start centre position to tool end centre position.
Hole Length = Routed Path Length + Tool Size = Slot length as defined in the PCB layout